

Application Deployment

(Deploy the given React application to a production ready state)

Step:1 Create a EC2 Instance

- Ubuntu
- T2.medium

Step: 2 Clone the Repository

```
ubuntu@ip-172-31-91-30:~$ git --version
git version 2.53.0
```

```
git version 2.53.0
ubuntu@ip-172-31-91-30:~$ git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 11 (delta 0), reused 0 (delta 0), pack-reused 10 (from 1)
Receiving objects: 100% (11/11), 101.89 KiB | 7.28 MiB/s, done.
ubuntu@ip-172-31-91-30:~$ cd Brain-Tasks-App
```

```
ubuntu@ip-172-31-91-30:~$ cd Brain-Tasks-App
ubuntu@ip-172-31-91-30:~/Brain-Tasks-App$ ls
README.md  dist
```

Step: 3 Install docker

```
Run 'docker --help' for more information
ubuntu@ip-172-31-91-30:~/Brain-Tasks-App$ docker --version
Docker version 29.6.2, build dfc4efb
```

```
total 16
-rw-rw-r-- 1 ubuntu ubuntu 1710 Jul 17 06:38 README.md
drwxrwxr-x 3 ubuntu ubuntu 4096 Jul 17 06:38 dist
-rw-rw-r-- 1 ubuntu ubuntu 61 Jul 21 08:40 dockerfile
-rw-rw-r-- 1 ubuntu ubuntu 94 Jul 17 06:43 package-lock.json
```

Step: 4 Install kubectl

```
144 history
ubuntu@ip-172-31-91-30:~$ kubectl version --client
Client Version: v1.36.2
Kustomize Version: v5.8.1
```

Step: 5 YAML files

```
-rw-rw-r-- 1 ubuntu ubuntu 1710 Jul 17 06:38 README.md
-rw-rw-r-- 1 ubuntu ubuntu 134 Jul 21 09:13 curl_response
-rw-rw-r-- 1 ubuntu ubuntu 450 Jul 21 09:59 deployment.yaml
drwxrwxr-x 3 ubuntu ubuntu 4096 Jul 17 06:38 dist
-rw-rw-r-- 1 ubuntu ubuntu 61 Jul 21 08:40 myapp.Dockerfile
-rw-rw-r-- 1 ubuntu ubuntu 94 Jul 17 06:43 package-lock.json
-rw-rw-r-- 1 ubuntu ubuntu 207 Jul 22 04:57 service.yaml
```

```
ubuntu@ip-172-31-91-30:~/Brain-Tasks-App$ ls -l
total 32
-rw-rw-r-- 1 ubuntu ubuntu 1710 Jul 17 06:38 README.md
-rw-rw-r-- 1 ubuntu ubuntu 971 Jul 22 05:06 buildspec.yml
-rw-rw-r-- 1 ubuntu ubuntu 134 Jul 21 09:13 curl_response
-rw-rw-r-- 1 ubuntu ubuntu 450 Jul 21 09:59 deployment.yaml
drwxrwxr-x 3 ubuntu ubuntu 4096 Jul 17 06:38 dist
-rw-rw-r-- 1 ubuntu ubuntu 61 Jul 21 08:40 myapp.Dockerfile
-rw-rw-r-- 1 ubuntu ubuntu 94 Jul 17 06:43 package-lock.json
-rw-rw-r-- 1 ubuntu ubuntu 200 Jul 22 05:03 service.yaml
```

Step: 6 Deploy these files

```
ubuntu@ip-172-31-91-30:~/Brain-Tasks-App$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/brain-tasks-app unchanged
service/brain-tasks-service unchanged
```

Step: 7 Buildcode from AWS

```
ubuntu@ip-172-31-91-30:~/Brain-Tasks-App$ aws codebuild import-source-credentials \
  --server-type GITHUB \
  --auth-type PERSONAL_ACCESS_TOKEN \
  --token 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
{
  "arn": "arn:aws:codebuild:us-east-1:112284275331:token/github"
```

Project created
You have successfully created the following project: brain-tasks-build
[Create a notification rule for this project](#)

brain-tasks-build

Actions ▼ Create trigger Edit Clone Debug build Start build with overrides Start build

This project uses Amazon Linux 2 as the host operating system. CodeBuild will continue to support Amazon Linux 2, but we recommend that you move to Amazon Linux 2023. To change it, in the Edit project Environment section under Additional configuration, choose Amazon Linux 2023 for Host kernel.
[Learn more](#)

Configuration

Source provider GitHub	Primary repository Sagarmal47/Final-Project-1	Artifacts upload location -	Service role arn:aws:iam::112284275331:role/service-role/codebuild-brain-tasks-build-service-role
Public builds			

Step: 8 Now add the Policy IAM Roles

Permissions policies (6) [Info](#)

[Simulate](#) [Remove](#) [Add permissions](#)

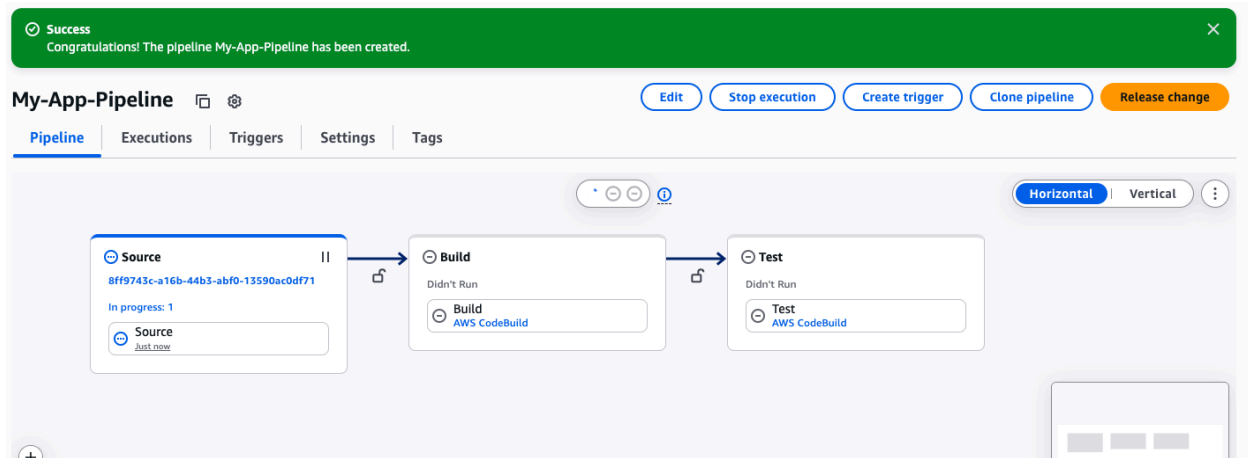
You can attach up to 10 managed policies.

Filter by Type
All types ▼

< 1 > [Settings](#)

☐	Policy name ↗	Type	Attached en... ▼
☐	AmazonEC2ContainerRegistryFullAccess	AWS managed	1
☐	AmazonEKSClusterPolicy	AWS managed	4
☐	AmazonEKSServicePolicy	AWS managed	1
☐	AmazonS3FullAccess	AWS managed	2
☐	CloudWatchLogsFullAccess	AWS managed	1
☐	CodeBuildBasePolicy-brain-tasks-build-us-east-1	Customer managed	1

Step: 9 Amazon Codepipeline



Step: 10 Verify the Deployment

```
ubuntu@ip-172-31-91-30:~/Brain-Tasks-App$ kubectl get deployments
kubectl get pods
kubectl get svc
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
brain-tasks-app  0/2     2            0           70m
NAME      READY   STATUS    RESTARTS   AGE
brain-tasks-app-659f6b75df-7p4g9  0/1     Pending   0           70m
brain-tasks-app-659f6b75df-hn6hp  0/1     Pending   0           70m
NAME      TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
brain-tasks-service  LoadBalancer  10.100.75.152  a1f2f1fbb4e6d4f0393792327ca3fbb9-1844728308.us-east-1.elb.amazonaws.com  80:31327/TCP    68m
kubernetes  ClusterIP     10.100.0.1    <none>        443/TCP          20h
```

Step: 11 kubectl get svc

```
ubuntu@ip-172-31-91-30:~/Brain-Tasks-App$ kubectl get svc
NAME      TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
brain-tasks-service  LoadBalancer  10.100.75.152  a1f2f1fbb4e6d4f0393792327ca3fbb9-1844728308.us-east-1.elb.amazonaws.com  80:31327/TCP    69m
kubernetes  ClusterIP     10.100.0.1    <none>        443/TCP          20h
```